

Boundary Contact, Residue Data, and the Original Arithmetic Trace

Split-Zero research programme

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Adjoining the full zero semilattice gives a generalized reciprocal at supported zero. Applying that reciprocal to labelled-root dynamics creates a definite collision-contact term. This paper computes its full analytic-unit expression and its image in the derivative of the original arithmetic trace. The same term is carried into fixed Cauchy and Laurent tests, with the original heat time retained.

The second calculation identifies the exact residue data needed for the earlier fixed-test correction and determinant. A double-zero algebra misses one unit coefficient; its next analytic thickening supplies it, and the quotient and transpose maps are proved explicitly. The final calculation independently reproduces the supplied nineteen- coefficient arithmetic witness from complete theta integrals with directed interval bounds. Its value is negative at time -2 and positive at time 0 .

The supplied `BOUNDARY_DEFECTS_AND_ARITHMETIC_RETURN.tex` is preserved unchanged alongside this paper and credited where used. The complete earlier HR, HD/HF/HL and HC/HH sources are also retained. Every additional result below has its full proof. A local contact contribution is not asserted to be a negative value of the complete original Weil form. No new arithmetic zero multiplicity is assigned.

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1 The full-unit contact defect and its arithmetic trace derivative

The supplied *Regular-inverse contact defects and their arithmetic return*, Theorem `thm:contact` and Proposition `prop:contactcotangent`, proves the finite labelled contact forcing and its nonzero perfect-residue value. Its complete original LaTeX accompanies this continuation. We retain that result and compute its receiver in the derivative of the original zero trace, with every analytic-unit term. This is a comparison of two specified evolutions; it does not change the definition of the arithmetic heat family. The human heat source is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, original equations `hoz`, `htdef`, `sas`. The original residue/trace conventions are proved in HH23–28.

BT1. The specified trajectories retain the environment. Let I be a real time interval, $x_1, \dots, x_n : I \rightarrow \mathbb{C}$ locally absolutely continuous, and $U(t, z)$ a nonvanishing function on an open spatial neighbourhood of all trajectories. Assume U is C^1 in real time and holomorphic in z , with the displayed spatial derivatives continuous locally. Put

$$P = \prod_{i=1}^n (z - x_i), \quad F = UP, \quad g = \frac{U_z}{U}, \quad \iota(a) = \begin{cases} a^{-1}, & a \neq 0, \\ 0, & a = 0. \end{cases} \quad (\text{BT1})$$

We compute on the exact extension with all labels retained:

$$\dot{x}_i = 2 \sum_{j \neq i} \iota(x_i - x_j) + 2g(t, x_i) \quad \text{almost everywhere.} \quad (\text{BT2})$$

This includes the original unit drift. It is the ordinary simple-root equation of $F_t = -F_{zz}$ when the roots are distinct and that PDE holds, by differentiation at each root. It specifies the additional rule at coincidence and therefore is not an assertion that the original equation has prescribed such a rule.

Define $P_i = \prod_{j \neq i} (z - x_j)$ and $P_{ij} = \prod_{\ell \neq i, j} (z - x_\ell)$. The complete defect is

$$\begin{aligned} D(t, z) &= 2 \sum_{i < j} \mathbf{1}_{x_i = x_j} P_{ij}, \\ B(t, z) &= \frac{U_t + U_{zz}}{U} + 2 \sum_i \frac{g(t, z) - g(t, x_i)}{z - x_i}, \end{aligned} \quad (\text{BT3})$$

$$\boxed{F_t + F_{zz} = UD + FB.}$$

Each divided difference in B is holomorphic at $z = x_i$, with value $g_z(t, x_i)$, including coincident labels. To prove the identity, pair the reciprocal terms in $P_t = -\sum_i \dot{x}_i P_i$. The polynomial identity $P_i - P_j = (x_i - x_j)P_{ij}$ gives

$$P_t + P_{zz} = D - 2 \sum_i g(t, x_i) P_i. \quad (\text{BT4})$$

Now expand $(\partial_t + \partial_z^2)(UP)$ and use $P_z = \sum_i P_i$. The remaining terms are $(U_t + U_{zz})P + 2U \sum_i (g(t, z) - g(t, x_i))P_i = FB$, using $(z - x_i)P_i = P$. This proves BT3 coefficientwise almost everywhere and distributionally on compact time intervals. All terms are locally integrable: the polynomial coefficients and U have the stated regularity, and D is a finite sum of bounded measurable coefficient functions.

BT2. No analytic unit cancels the contact class. At a fixed time let $c_1, \dots, c_r$ be the distinct centres and $m_1, \dots, m_r$ their multiplicities. Then

$$D = \sum_{l: m_l \geq 2} m_l(m_l - 1)(z - c_l)^{m_l - 2} \prod_{h \neq l} (z - c_h)^{m_h}, \quad \frac{D}{P} = \sum_{l: m_l \geq 2} \frac{m_l(m_l - 1)}{(z - c_l)^2}. \quad (\text{BT5})$$

The second identity is meromorphic and exact, not an asymptotic truncation. In the local coordinate $x = z - c_l$, put $u(x) = U(t, c_l + x) \prod_{h \neq l} (c_l + x - c_h)^{m_h}$. It is the full original local unit of $F = x^{m_l} u(x)$. Modulo x^{m_l} the defect in BT3 is

$$q_l = m_l(m_l - 1)x^{m_l-2}u(x). \quad (\text{BT6})$$

All other contact summands and FB vanish in that local algebra. The leading coefficient in BT6 is nonzero for $m_l \geq 2$. Consequently $F_t + F_{zz} = 0$ forces every collision set to have measure zero; the analytic environment term cannot cancel contact. The exact converse keeps the environment: the PDE holds if and only if $D = 0$ and $B = 0$ almost everywhere. Indeed if the PDE holds, BT6 forces $D = 0$ at almost every time; BT3 then gives $FB = 0$ and hence $B = 0$ by holomorphy. Conversely both zero terms imply the PDE. When $U = 1$, $B = 0$ identically and this recovers precisely the source's residence criterion. No converse that discards U is used.

BT3. The residue detects a derivative that the value trace can kill. At one centre of multiplicity m retain

$$A = \mathbb{C}[x]/(x^m), \quad F_0 = x^m u(x), \quad \lambda_u(a) = \text{Res}_{x=0} \frac{a(x)}{x^m u(x)} dx. \quad (\text{BT7})$$

For a spatially holomorphic test ϕ , consider the infinitesimal coefficient perturbation $F_0 + \varepsilon Q$, with $\varepsilon^2 = 0$ and Q holomorphic. Differentiating the fixed root-isolating contour gives the exact zero-trace variation

$$\mathcal{T}_\phi([Q]) = \text{Res}_0 \phi \partial_x (Q/F_0) dx = -\lambda_u(\phi' Q). \quad (\text{BT8})$$

It depends only on $[Q] \in A$: adding $F_0 V$ adds the derivative of a holomorphic function and contributes no residue. The contour formula follows by differentiating F_x/F on the contour, where F_0 is nonzero, and integration by parts of a meromorphic differential proves the second equality. It does not require choosing derivatives of colliding roots.

For the contact class $q = m(m-1)x^{m-2}u$ in BT6,

$$\boxed{\lambda_u(xq) = m(m-1), \quad \mathcal{T}_\phi(q) = -m(m-1)\phi''(0).} \quad (\text{BT9})$$

The full unit cancels in these particular quotients; it remains in q , F_0 , λ_u and BT8. Since $q/F_0 = m(m-1)/x^2$, the first formula is a simple residue and the second follows by extracting the linear coefficient of ϕ' .

For $m \geq 3$, the original cotangent complex and its duality are

$$[A \xrightarrow{F'_0} A dx], \quad H^{-1} = (x), \quad H^0 = A/(F'_0) dx, \quad \langle a, [b dx] \rangle = \lambda_u(ab). \quad (\text{BT10})$$

Here $(F'_0) = (x^{m-1})$ since $u(0) \neq 0$. The residue pairing on A is perfect: its matrix in the full power basis is anti-triangular with nonzero anti-diagonal $1/u(0)$. The orthogonal complement of (F'_0) is its annihilator, proving the perfect induced pairing in BT10. Thus

$$q \in H^{-1}, \quad \mathcal{T}_\phi(q) = -\langle q, d\phi \rangle, \quad \text{Tr}_A(M_{qa}) = 0 \quad (a \in A). \quad (\text{BT11})$$

This is the exact connecting map from the trace-radical contact class to a measurable trace derivative. At $m = 2$ the class is $q = 2u$, its value trace is $4u(0)$, and $F'_0 q = 4u(0)^2 x \neq 0$ in A . Formula BT9 still holds, but the H^{-1} assertion is not made in that case.

BT4. Coordinate changes and the exact test restriction. The variation BT8 is defined on the entire marked coefficient space A . A coordinate-change direction $Q = F'_0 v$ has

$$\mathcal{T}_\phi(F'_0 v) = -m\phi'(0)v(0). \quad (\text{BT12})$$

This follows from $F'_0/F_0 = m/x + u'/u$ by taking its residue. Therefore BT8 descends to the quotient $A/(F'_0)$ exactly when $\phi'(0) = 0$ (for $m \geq 2$). When $\phi'(0) \neq 0$, taking $v = 1$ exhibits the failure and computes its full correction. The quotient has thus not been substituted for the original marked deformation. For reflected products of tests that both vanish at the collision, the derivative condition holds automatically; this is the test class in which the contact pairing descends.

BT5. The exact arithmetic coefficient and its sign. Use the original arithmetic variable

$$s = \rho + ix/2, \quad \rho = 1/2 + ic_l/2, \quad G(t, s) = 16F(t, -2i(s - 1/2)), \quad \mathcal{E} = \partial_t G - \frac{1}{4}G_{ss}. \quad (\text{BT13})$$

The constants follow from $(-2i)^2 = -4$. Consequently $\mathcal{E} = 16(F_t + F_{zz})$ and

$$\partial_t(G_s/G) = \frac{1}{4}\partial_s(G_{ss}/G) + \partial_s(\mathcal{E}/G), \quad \frac{\mathcal{E}}{G} = -\sum_l \frac{m_l(m_l - 1)}{4(s - \rho_l)^2} + B(t, -2i(s - 1/2)). \quad (\text{BT14})$$

All terms involving the environment in the second summand are holomorphic on the retained cluster neighbourhoods. Thus for any fixed holomorphic arithmetic test $A(s)$, the contact contribution to the actual derivative of the finite zero trace is

$$\boxed{\delta_{\text{contact}} \dot{Z}(A) = \sum_l \frac{m_l(m_l - 1)}{4} A''(\rho_l)}. \quad (\text{BT15})$$

This is BT9 with $\phi(x) = A(\rho + ix/2)$, or directly the residue of $A\partial_s(\mathcal{E}/G)$ in BT14. The factor $1/4$ and its sign are forced by the original arithmetic coordinate.

If the centre is real in the spatial coordinate, $\rho = 1 - \bar{\rho}$. For the original involution $F^\#(s) = \overline{F(1 - \bar{s})}$, take two arithmetic tests F, G vanishing at ρ and put $A = F^\#G$. Direct differentiation gives $A''(\rho) = -2\overline{F'(\rho)}G'(\rho)$. Hence

$$\delta_{\text{contact}} \dot{Z}(F^\#G) = -\frac{m(m - 1)}{2} \overline{F'(\rho)}G'(\rho). \quad (\text{BT16})$$

This is an evaluated negative first-jet response of the added contact term. The genuine coefficient heat flow has the opposite positive first-jet response. To prove this last comparison with the complete unit, write $G_0 = (s - \rho)^m v(s)$; the principal part of $G''_0/(4G_0)$ is

$$\frac{m(m - 1)}{4(s - \rho)^2} + \frac{m}{2} \frac{v'(\rho)}{v(\rho)} \frac{1}{s - \rho}. \quad (\text{BT17})$$

Differentiate in s and take the residue against A . The true heat variation is

$$\dot{Z}_{\text{heat}, \rho}(A) = -\frac{m(m - 1)}{4} A''(\rho) - \frac{m}{2} \frac{v'(\rho)}{v(\rho)} A'(\rho). \quad (\text{BT18})$$

Adding BT15 cancels its first term exactly. The retained unit drift remains. For the vanishing test pair, $A'(\rho) = 0$ also, so the totalized persistent-contact flow has zero derivative on that first-jet test, whereas the genuine heat family has the positive value. This explicitly measures the changed dynamics.

BT6. Fixed original Cauchy tests observe the full contact matrix. Let z_a, z_b be distinct or equal poles with real parts greater than one, $F_z(s) = 1/(s - z)$, and $A_{ab} = F_{z_a}^\# F_{z_b}$. Put $\alpha = 1 - \bar{z}_a$, $\beta = z_b$, and for a centre ρ_l put $d_l = \rho_l - \alpha$, $e_l = \rho_l - \beta$. As these poles lie outside the original critical strip, they do not meet an original arithmetic zero. In any retained local continuation they are kept away from the cluster. Then

$$A''_{ab}(\rho_l) = -2 \left(\frac{1}{d_l^3 e_l} + \frac{1}{d_l^2 e_l^2} + \frac{1}{d_l e_l^3} \right), \quad (\text{BT19})$$

and the exact contact matrix is

$$(\delta_{\text{contact}} \dot{C})_{ab} = - \sum_l \frac{m_l(m_l - 1)}{2} \left(\frac{1}{d_l^3 e_l} + \frac{1}{d_l^2 e_l^2} + \frac{1}{d_l e_l^3} \right). \quad (\text{BT20})$$

Twice differentiating $A_{ab}(s) = -1/((s - \alpha)(s - \beta))$ proves BT19 and BT15 gives BT20. The original fixed-test frame HR7 maps this matrix into the full collision algebra; no test coefficient is discarded. On its subspace of tests vanishing at a critical collision, the single-centre matrix reduces to the rank-one negative form BT16. This remains a contribution to the changed-flow derivative, not a negative value of the complete original arithmetic form at zero.

BT7. Exact return to the genuine equation and complete support. For $U = 1$ the source's finite return map is

$$P(t) = e^{-(t-t_0)\partial_z^2} P(t_0) + \int_{t_0}^t e^{-(t-r)\partial_z^2} D(r) dr. \quad (\text{BT21})$$

The exponential terminates on degree- n polynomials, and differentiating after the inverse exponential proves BT21 coefficientwise for absolutely continuous paths. For the actual unit the complete returned equation is BT3, including B ; there is no omitted external-root contribution.

Finally the input inverse is exactly the generalized inverse in the original $G_L(\mathbb{C})$. In its top supported ring fibre,

$$d(a) = 1 - aa^\dagger, \quad p(d(a)) = \mathbf{1}_{p(a)=0}, \quad p(D) = 2 \sum_{i < j} p(d(x_i - x_j)) P_{ij}. \quad (\text{BT22})$$

Here differences are taken in the top supported fibre, whose additive identity is e . Thus the contact indicator detects supported equality and has never identified it with absence τ . The linear map $Q \mapsto \mathcal{T}_\phi(Q)$ lifts as a $G_L(\mathbb{C})$ -semimodule map on the full supported carriers: $(Q, 1_L) \mapsto (\mathcal{T}_\phi(Q), 1_L)$ and $(0, \lambda) \mapsto (0, \lambda)$. Linearity of BT8 preserves amplitude addition and scalar action; unchanged labels preserve support joins and scalar meets. The residue pairings lift as bilinear maps, with output support the meet of the input supports. Distributivity of L verifies both additive laws, and residue bilinearity verifies the amplitudes. Zero amplitudes retain their declared support labels. In particular the zero value trace of BT11 remains a supported zero, while BT9 gives a nonzero tangent observation on the same nonzero class.

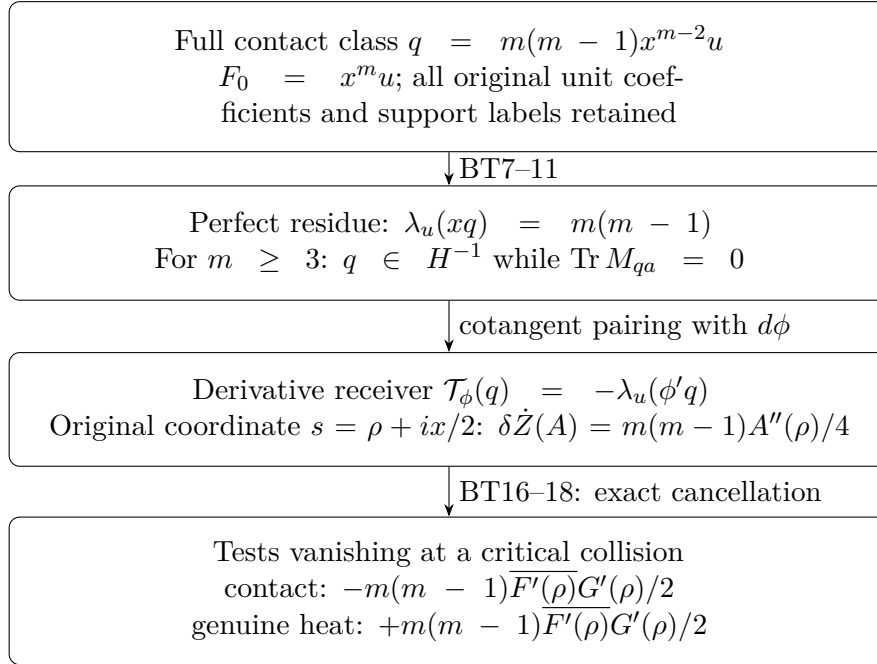


Figure 1: The contact class has a nonzero trace-derivative receiver even when its ordinary trace pairing vanishes. Its added derivative cancels the genuine heat splitting term. These are two computed contributions to the specified equations; their difference retains the original clock, analytic unit and arithmetic tests. Complete proofs BT1-22; the supplied contact derivation and human heat source are cited above.

2 The sharp residue receiver for fixed arithmetic heat tests

The supplied *Regular-inverse contact defects and their arithmetic return*, Theorem `thm:residuejet`, proves the three-coefficient receiver for collision energy and exhibits the missing coefficient at a double collision. The result here carries that coefficient into every fixed-test layer HR4 and the local determinant correction HR14 of `FIXED_ARITHMETIC_HEAT_FLAGS.tex`. The full flag source `HEAT_TRACE_FLAG_RESPONSE.tex`, HF1–16, is also retained. Those two complete local sources and the supplied theorem were read. The heat convention is Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, author equations `hoz`, `htdef`, and `sas`, as identified in HR and HF. This component does not claim an independent reading of their entire paper. All receiver maps, thresholds, and finite heat examples used below are proved here.

RJ1. Original coordinates and marked data. Let $H(t, x_0 + x)$ be a jointly holomorphic solution of $\partial_t H = -\partial_x^2 H$, of real type for real t, x , with an actual zero of order $m \geq 1$ at $(0, x_0)$. Retain

$$H(0, x_0 + x) = x^m u(x), \quad u(x) = u_0 + u_1 x + u_2 x^2 + \cdots, \quad u_0 \neq 0, \quad \rho = \frac{1}{2} + \frac{ix_0}{2}, \quad s = \rho + \frac{ix}{2}. \quad (\text{RJ1})$$

In particular the actual arithmetic family may be used, with $H_0(z) = \xi_{\mathbb{R}}(1/2 + iz/2)/8$. No multiplicity of an actual arithmetic zero is assigned by the following construction. Write

$$c = \frac{u_1}{u_0}, \quad d = \frac{u_2}{u_0}, \quad b = 2d - c^2 = (\log u)''(0). \quad (\text{RJ2})$$

Thus c, d, b are exactly the HR1 coefficients. The letter b in RJ2 is not the letter b used for the coefficient u_2/u_0 in the supplied energy paper: that paper's a, b, κ are respectively c, d, b here. We keep the entire u , not a substituted polynomial unit, when defining the receiver.

For $n \geq 1$ define a coordinate-marked algebra and functional by

$$A_n = \mathbb{C}[x]/(x^n), \quad \lambda_n(f) = \text{Res}_{x=0} \frac{f(x)}{x^n u(x)} dx, \quad B_n(f, g) = \lambda_n(fg). \quad (\text{RJ3})$$

Here “marked” specifies the class of the original x , the identity, and the given functional; arbitrary changes of coordinate are not identified with the same observation. The functional is independent of the polynomial representative: adding $x^n h$ changes its integrand by the holomorphic function h/u . The actual collision receiver is A_m . Other values of n record analytic thickenings or quotients, not a changed zero multiplicity or a changed heat clock.

RJ2. What the entire residue pairing records. Put $u(x)^{-1} = \sum_{k \geq 0} q_k x^k$. Direct extraction of the coefficient of x^{-1} gives

$$\lambda_n(x^a) = q_{n-1-a} \quad (0 \leq a < n), \quad B_n(x^a, x^b) = \begin{cases} q_{n-1-a-b}, & a+b < n, \\ 0, & a+b \geq n. \end{cases} \quad (\text{RJ4})$$

Reversing the columns of this matrix makes it triangular with diagonal $q_0 = u_0^{-1}$. Therefore

$$\det B_n = (-1)^{n(n-1)/2} u_0^{-n} \neq 0. \quad (\text{RJ5})$$

This proves perfection with its full matrix, including all coefficients of u that can enter at this order. Conversely, $B_n(1, f) = \lambda_n(f)$, so knowing the entire marked pairing gives exactly $q_0, \dots, q_{n-1}$. The identities $u(x) \sum q_k x^k = 1$ recover

$$u_0 = q_0^{-1}, \quad u_k = -q_0^{-1} \sum_{i=1}^k q_i u_{k-i} \quad (1 \leq k < n). \quad (\text{RJ6})$$

Hence two units give the same marked receiver at order n if and only if their jets agree modulo x^n . This is an exact equivalence; perfection alone does not add any unobserved coefficient.

RJ3. Every comparison map in the residue tower. For $N = n + r$, $r \geq 1$, let $\pi_{N,n} : A_N \rightarrow A_n$ be the quotient algebra map, and define

$$j_{n,N}(f) = x^r \tilde{f} \pmod{x^N}, \quad 0 \longrightarrow (x^n)/(x^N) \longrightarrow A_N \xrightarrow{\pi_{N,n}} A_n \longrightarrow 0. \quad (\text{RJ7})$$

A different lift changes $x^r \tilde{f}$ by a multiple of x^N , so j is well defined. Its image is (x^r) and its kernel is zero, because the coefficients of $x^r, \dots, x^{N-1}$ are precisely the original n coefficients. The kernel of π is (x^n) , of dimension r , identified with A_r as a module by multiplication by x^n . Give A_n the A_N action through π . Then j is an A_N -module map, since $ax^r \tilde{f} = x^r \pi(a) \tilde{f}$ modulo x^N . For $r > 0$ it is not a unital algebra map: it sends 1 to x^r .

The full adjunction identity is

$$\boxed{\lambda_n(f) = \lambda_N(j_{n,N}f), \quad B_n(\pi_{N,n}a, f) = B_N(a, j_{n,N}f).} \quad (\text{RJ8})$$

Both statements follow from cancellation of x^r in the residue denominator; the second also follows from the module identity. Thus under the perfect dualities B_N, B_n , the transpose of the quotient map is exactly j . In particular

$$(\ker \pi_{N,n})^{\perp_{B_N}} = \text{im } j_{n,N} = (x^r). \quad (\text{RJ9})$$

The inclusion follows from RJ8. Both sides have dimension n by perfection and the already computed kernel dimension, proving equality. Composition retains the whole tower: $\pi_{N,n}\pi_{P,N} = \pi_{P,n}$ and $j_{N,P}j_{n,N} = j_{n,P}$ for $P \geq N \geq n$. These formulas are literal quotient composition and multiplication by $x^{P-N}x^{N-n}$.

All support labels can be retained at these maps. On the original full-support carrier $\{(0, \ell) : \ell \in L\} \cup \{(a, 1_L) : a \in A_n\}$, a linear map ϕ above sends $(a, 1_L)$ to $(\phi(a), 1_L)$ and fixes every $(0, \ell)$. This is well defined on the common top-supported zero. It preserves addition because amplitudes add and labels join. Scalar action is preserved with the module actions specified above. A nonzero element in the kernel of π becomes $(0, 1_L)$, retaining its top label; it does not become external absence. The quotient is also multiplicative, since π is an algebra map and labels meet. Only the asserted module structure is preserved by j . The pairings in RJ8 retain the same output support on both sides: the input labels have not changed, including when the scalar residue is zero.

RJ4. The least residue orders for the original unit. For every $n \geq 3$ write

$$L_0 = \lambda_n(x^{n-1}), \quad L_1 = \lambda_n(x^{n-2}), \quad L_2 = \lambda_n(x^{n-3}), \quad A = \frac{L_1}{L_0}, \quad B = \frac{L_2}{L_0}. \quad (\text{RJ10})$$

The same L_0, L_1 exist at $n = 2$. Equations RJ4–6 give

$$L_0 = u_0^{-1}, \quad L_1 = -cu_0^{-1}, \quad L_2 = (c^2 - d)u_0^{-1}, \quad \boxed{c = -A, \quad d = A^2 - B, \quad b = A^2 - 2B.} \quad (\text{RJ11})$$

The tower proves these are independent of the chosen receiver order once that order is large enough. Order 2 is the least order that determines c for all units: $1 + \alpha x$ has fixed order-1 receiver and arbitrary $c = \alpha$. Order 3 is the least order that determines either d or b : $1 + cx + dx^2$ has the same order-2 receiver as d varies, but both d and $b = 2d - c^2$ vary. These are comparisons of whole functionals and whole perfect pairings, not selected entries alone. RJ9 describes the exact duality map between these objects; no inference from a quotient's nonidentity is used to deny their relation.

RJ5. Heat expansion in the original time. We give the analytic calculation needed to apply this receiver, including the constants in HF and HR. Let

$$P_k(y) = \sum_{a=0}^{\lfloor k/2 \rfloor} \frac{(-1)^a k!}{a!(k-2a)!} y^{k-2a}. \quad (\text{RJ12})$$

Coefficient differentiation gives $P'_k = kP_{k-1}$, $P_{k+1} = yP_k - 2kP_{k-1}$, and $2P''_k - yP'_k + kP_k = 0$. All roots of P_k are real and simple. One proof uses $P_k = (-2)^k e^{y^2/4} (d/dy)^k e^{-y^2/4}$, obtained by coefficient differentiation. Integration by parts k times makes P_k orthogonal to every polynomial of degree less than k against $e^{-y^2/4} dy$; every boundary term vanishes by Gaussian decay. If P_k had fewer than k sign-changing real roots, multiply the linear factors at those roots. The resulting polynomial has degree less than k , and its product with P_k has one sign, is not zero, and is integrable against that strictly positive weight. Its integral cannot vanish, a contradiction. Thus there are exactly k distinct real roots. Denote those of P_m by $r_1, \dots, r_m$, and put $p_a = \sum_{\nu} r_{\nu}^a$, including $p_0 = m$.

Set $t = \epsilon^2$, $x = \epsilon y$ for the root calculation only. The PDE forces $\partial_t^a \partial_x^k H(0, x_0) = (-1)^a \partial_x^{k+2a} H(0, x_0)$. In the convergent two-variable Taylor series, every weighted degree $k+2a < m$ therefore vanishes. Grouping the finitely many terms at each of the next three weighted degrees gives the analytic identity

$$\frac{H(\epsilon^2, x_0 + \epsilon y)}{u_0 \epsilon^m} = P_m(y) + c\epsilon P_{m+1}(y) + d\epsilon^2 P_{m+2}(y) + O(\epsilon^3). \quad (\text{RJ13})$$

The remainder is holomorphic in (ϵ, y) near every bounded y -set; division by ϵ^m is legitimate because the lower coefficients vanished. This uses the given joint analyticity and does not assume convergence of an unrestricted heat series.

At a root r the recurrences give $P_{m+1}(r) = -2P'_m(r)$, $P'_{m+1}(r) = 0$, $P_{m+2}(r) = -2rP'_m(r)$, and $P''_m(r)/P'_m(r) = r/2$. The implicit function theorem applied to RJ13 and these four identities gives the first two root corrections $2c$ and $(2d - c^2)r$. The complete local cluster consequently has

$$\theta_{\nu}(\epsilon) = r_{\nu}\epsilon + 2c\epsilon^2 + br_{\nu}\epsilon^3 + O(\epsilon^4). \quad (\text{RJ14})$$

To verify completeness, take a small fixed disk in x containing only the zero at 0 of the initial germ. On its boundary H_0 is nonzero, and uniform continuity and the argument principle keep the number of zeros equal to m for small t . The m distinct branches just constructed exhaust them. Changing ϵ to $-\epsilon$ leaves their unordered set unchanged. Thus every symmetric holomorphic expression in them is analytic in the original t . In particular, with $s_a(t) = \sum_{\nu} \theta_{\nu}^a$,

$$s_{2j}(t) = t^j [p_{2j} + t(2jb p_{2j} + 4j(2j-1)c^2 p_{2j-2}) + O(t^2)], \quad j \geq 1, \quad (\text{RJ15})$$

$$s_{2j+1}(t) = 2(2j+1)cp_{2j}t^{j+1} + O(t^{j+2}), \quad j \geq 0, \quad (\text{RJ16})$$

$$s_{2j+2}(t) = p_{2j+2}t^{j+1} + O(t^{j+2}), \quad j \geq 0. \quad (\text{RJ17})$$

The coefficient at degree $2j+2$ in RJ15 comes from either one $br_{\nu}\epsilon^3$ or two $2c\epsilon^2$ factors, giving the two displayed terms. The analogous single $2c\epsilon^2$ choice gives RJ16. All odd root moments vanish by $P_m(-y) = (-1)^m P_m(y)$. The next possible odd error powers disappear because the sums are even in ϵ , proving the stated errors in t . Also $s_0 = m$ exactly. Newton's identity gives $p_2 = 2m(m-1)$, and $p_{2j} > 0$ for $m \geq 2$, $j \geq 1$, since at least one real root is nonzero.

RJ6. The residue formula for every fixed-test layer. Let F, G be fixed holomorphic functions near ρ , with $f_a = F^{(a)}(\rho)/a!$, $g_a = G^{(a)}(\rho)/a!$, vanishing for $a < j$, where $0 \leq j < m$. Reflection is $F^{\#}(s) = \overline{F(1 - \bar{s})}$. In the coordinate RJ1, the pullback coefficients are $(i/2)^a f_a$ and

$(i/2)^a g_a$; reflection conjugates their coefficients. Consequently the coefficients of their product at $x^{2j}, x^{2j+1}, x^{2j+2}$ are

$$\frac{\bar{f}_j g_j}{4^j}, \quad \frac{i(\bar{f}_j g_{j+1} - \bar{f}_{j+1} g_j)}{2 \cdot 4^j}, \quad \frac{\bar{f}_{j+1} g_{j+1} - \bar{f}_j g_{j+2} - \bar{f}_{j+2} g_j}{4^{j+1}}. \quad (\text{RJ18})$$

Insert RJ15–17 into the finite branch sum $Z_{\rho,t}(F^\# G) = \sum_\nu (F^\# G)(\rho + i\theta_\nu/2)$. The convergent remaining Taylor series is $O(\epsilon^{2j+3})$ on each branch, and its symmetric sum is even in ϵ ; it is therefore $O(t^{j+2})$. We have proved

$$Z_{\rho,t}(F^\# G) = \frac{t^j}{4^j} [p_{2j} \bar{f}_j g_j + t \mathcal{C}_j(F, G) + O(t^2)], \quad (\text{RJ19})$$

with the following residue expression for the entire HR4 coefficient:

$$\begin{aligned} \mathcal{C}_j(F, G) = & [(2j p_{2j} + 4j(2j-1)p_{2j-2}) A^2 - 4j p_{2j} B] \bar{f}_j g_j \\ & - i(2j+1) A p_{2j} (\bar{f}_j g_{j+1} - \bar{f}_{j+1} g_j) \\ & + \frac{p_{2j+2}}{4} (\bar{f}_{j+1} g_{j+1} - \bar{f}_j g_{j+2} - \bar{f}_{j+2} g_j). \end{aligned} \quad (\text{RJ20})$$

For $j = 0$ the entire first line is defined to be zero, with no negative-index moment. Only $A = L_1/L_0$ is then needed. For all $j \geq 1$, order 3 supplies both A, B . The test jets through degree $j+2$ in RJ20 are the actual fixed-test jets. They are not erased even if $j+2 \geq m$. The formula uses the full analytic unit through the exact coefficients it determines at this time order; it makes no claim that a finite receiver replaces u for higher orders.

RJ7. The residue formula for the fixed Cauchy determinant. Choose m distinct poles z_l with $\text{Re } z_l > 1$, put $a_l = \rho - z_l$, $F_l(s) = 1/(s - z_l)$, and let $C_\rho(t)_{al} = Z_{\rho,t}(F_a^\# F_l)$. With $M = m(m-1)/2$ the exact determinant is

$$\det C_\rho(t) = \text{disc } W(t, x) \frac{4^{-M} |\prod_{a < l} (z_l - z_a)|^2}{|\prod_{\nu, l} (a_l + i\theta_\nu/2)|^2} \quad \text{for real } t. \quad (\text{RJ21})$$

Here $W = \prod_\nu (x - \theta_\nu)$ is the original monic local factor, and its original analytic unit is retained in $H = UW$. For a proof, the trace matrix in the basis $1, x, \dots, x^{m-1}$ has entries s_{a+b} , so its determinant is the squared root Vandermonde. The coefficient columns of the Cauchy tests are the inverse denominators $(a_l I + (i/2)C_x)^{-1} e_0$. Evaluating them on the distinct branches gives the Cauchy matrix. Its determinant, after multiplication by its common denominator, is alternating in the row points and the poles; its degrees force the two Vandermonde factors. Induction on a successive simple-pole residue gives the constant $(-1)^M$. Dividing by the root evaluation Vandermonde therefore gives the coefficient determinant

$$\det R(t) = \frac{(-i/2)^M \prod_{a < l} (z_l - z_a)}{\prod_{\nu, l} (a_l + i\theta_\nu/2)}.$$

For real t , coefficient reflection gives $C_\rho = R^* K R$, including negative t ; it does not replace reflection by the Euclidean evaluation norm at nonreal roots. This proves RJ21 away from zero. Analytic continuation proves it at zero as well.

The squared Vandermonde in RJ14 has first relative correction $2Mb = m(m-1)b$. Its leading constant is

$$D_m = \prod_{a < l} (r_l - r_a)^2 = 2^M \prod_{k=1}^m k^k. \quad (\text{RJ22})$$

For completeness, $D_m = (-1)^M \prod_i P'_m(r_i)$ and $P'_m = m P_{m-1}$. The root-product identity for monic polynomials gives $\prod_i P_{m-1}(r_i) = \prod_j P_m(s_j)$, since $m(m-1)$ is even, where s_j are the

roots of P_{m-1} . The recurrence makes the right side $(-2(m-1))^{m-1} \prod_j P_{m-2}(s_j)$. Substitution in the two discriminant formulas gives $D_m = 2^{m-1} m^m D_{m-1}$; the signs cancel because $m-1 + M - (m-1)(m-2)/2 = 2(m-1)$. Starting at $D_1 = 1$ proves RJ22.

On a local logarithm at each nonzero a_l , RJ16 and p_2 give

$$\sum_{\nu} \log(a_l + i\theta_{\nu}/2) = m \log a_l + t \left(\frac{imc}{a_l} + \frac{m(m-1)}{4a_l^2} \right) + O(t^2).$$

The real part and its derivative are independent of the logarithm branch. We obtain the full HR13–14 expansion

$$\det C_{\rho}(t) = \frac{D_m 4^{-M} |\prod_{a < l} (z_l - z_a)|^2}{\prod_l |a_l|^{2m}} t^M (1 + \mathcal{E}_{\rho} t + O(t^2)), \quad (\text{RJ23})$$

with the exact residue correction

$$\mathcal{E}_{\rho} = m(m-1)(A^2 - 2B) + 2mA \operatorname{Re} \left(i \sum_l \frac{1}{a_l} \right) - \frac{m(m-1)}{2} \operatorname{Re} \sum_l \frac{1}{a_l^2}. \quad (\text{RJ24})$$

For $m = 1$ the terms with $m(m-1)$ vanish, so only A occurs. RJ24 preserves the pole configuration, every affine factor, and the original heat time.

RJ8. Genuine heat counterexamples prove every threshold. For every $m \geq 1$ and real c, d , define the finite polynomial

$$\mathbf{H}_{m;c,d}(t, x) = \sum_{a=0}^{\lfloor (m+2)/2 \rfloor} \frac{(-t)^a}{a!} \partial_x^{2a} [x^m (1 + cx + dx^2)]. \quad (\text{RJ25})$$

This is jointly entire, real for real arguments, and its initial zero at $x = 0$ has order exactly m . Termwise differentiation and reindexing prove $\partial_t \mathbf{H} = -\partial_x^2 \mathbf{H}$: the final unpaired derivative is zero because it exceeds the polynomial degree. Thus these are genuine analytic solutions in the actual clock, and their unit is exactly $1 + cx + dx^2$. The cluster completeness proof in RJ5 applies; no extra initial zeros of this polynomial are inserted into the multiplicity- m cluster. These examples test information sufficiency for analytic heat families. They are not asserted to be members of the Rodgers–Tao family.

First vary c with d fixed. The order-1 receiver is always $\lambda_1(1) = 1$. Taking the fixed tests $F(s) = 1$ and $G(s) = s - \rho$ in RJ20 gives $\mathcal{C}_0(F, G) = imc$. It changes with c , so order 1 does not determine the entire first fixed-test coefficient. Order 2 does by RJ20. The observation is Hermitian as well: take $F = G = 1 + i(s - \rho)$. Then $\mathcal{C}_0(F, F) = -2mc + p_2/4$, which also changes.

Next fix c and compare d with $d + \Delta$, $\Delta \neq 0$. The whole order-2 receiver is the same in both families: $\lambda_2(1) = -c$, $\lambda_2(x) = 1$ and $B_2 = \begin{pmatrix} -c & 1 \\ 1 & 0 \end{pmatrix}$. For every $1 \leq j < m$, with the same fixed tests in both families, RJ20 gives

$$\Delta \mathcal{C}_j(F, G) = 4jp_{2j} \Delta \bar{f}_j g_j, \quad \Delta \mathcal{E}_{\rho} = 2m(m-1)\Delta. \quad (\text{RJ26})$$

Choose $F = G = (s - \rho)^j$ to make the first change nonzero. This works for every $m \geq 2$ and every $1 \leq j < m$ because $p_{2j} > 0$. The determinant change is nonzero for every allowed pole configuration when $m \geq 2$. The universal order-3 sufficiency in RJ20 and RJ24 is therefore sharp.

These witnesses can also be taken inside a fixed Cauchy-test span. For the same poles define $D(s) = \prod_l (s - z_l)$ and

$$B_j(s) = \frac{(s - \rho)^j [D(\rho + v)]_{< m-j} |_{v=s-\rho}}{D(s)}. \quad (\text{RJ27})$$

The difference between its numerator and $(s - \rho)^j D(s)$ vanishes to order m , while $D(\rho) \neq 0$. Hence $B_j(s) = (s - \rho)^j + O((s - \rho)^m)$ and its leading jet is 1. Its numerator has degree at most $m - 1$, so subtracting its simple pole residues expresses it as a linear combination of the fixed $1/(s - z_l)$; the remaining rational function has no poles and vanishes at infinity, hence is zero. Its higher jets are fixed by these poles and stay unchanged when d varies. Thus RJ26 with $F = G = B_j$ remains $4jp_{2j}\Delta$ without dropping any of those jets. Every flag threshold is attained by the original fixed rational tests as well.

For $m = 1$, the determinant is the only layer-0 one-test Gram entry. Its correction equals $-2c \operatorname{Re}(i/a_1)$. It needs order 2 precisely when $\operatorname{Re}(i/a_1) \neq 0$, as varying c in RJ25 proves. Since $\operatorname{Re}(i/a_1) = \operatorname{Im}(a_1)/|a_1|^2$, it is identically zero at this time order when $\operatorname{Im} z_1 = \operatorname{Im} \rho$; then no unit derivative is required for this particular scalar coefficient. This exception does not change the order-2 threshold for the full layer-0 bilinear observation.

RJ9. What the extra datum changes at a double zero. For $m = 2$ the original A_2 already determines $c = -A$. The first larger receiver has the actual quotient and kernel

$$0 \longrightarrow \mathbb{C}x^2 \longrightarrow A_3 \xrightarrow{\pi_{3,2}} A_2 \longrightarrow 0, \quad \lambda_2(f) = \lambda_3(x\tilde{f}). \quad (\text{RJ28})$$

The new value is $L_2 = \lambda_3(1)$, while $L_1 = \lambda_3(x)$ and $L_0 = \lambda_3(x^2)$ were already recorded by A_2 . Since $p_0 = 2$, $p_2 = 4$, $p_4 = 8$, the complete layer-1 coefficient is

$$\begin{aligned} \mathcal{C}_1(F, G) = & 16(A^2 - B)\bar{f}_1g_1 - 12iA(\bar{f}_1g_2 - \bar{f}_2g_1) \\ & + 2(\bar{f}_2g_2 - \bar{f}_1g_3 - \bar{f}_3g_1), \end{aligned} \quad (\text{RJ29})$$

$$\mathcal{E}_\rho = 2A^2 - 4B + 4A \operatorname{Re} \left(i \sum_{l=1}^2 \frac{1}{a_l} \right) - \operatorname{Re} \sum_{l=1}^2 \frac{1}{a_l^2}. \quad (\text{RJ30})$$

Thus changing the new normalized datum by ΔB with the old receiver fixed changes these coefficients by $-16\Delta B \bar{f}_1g_1$ and $-4\Delta B$, respectively. The actual t^2 coefficient in $Z_{\rho,t}$ changes by $-4\Delta B \bar{f}_1g_1$, because RJ19 retains its factor $1/4$. The earlier energy paper already proved the insufficiency for its energy finite part. Equations RJ29–30 are its additional fixed-test and determinant receivers, with the full higher test jets included.

For an actual simple zero, A_1 lacks the coefficient c ; A_2 supplies it for the full first variation. For $m \geq 3$ the original A_m already contains the three required residue values, so every displayed fixed-test correction is read directly from that original receiver. Higher values of n are still related by RJ7–9. None of these statements changes m or turns a local correction into a sign of the complete arithmetic Weil form.

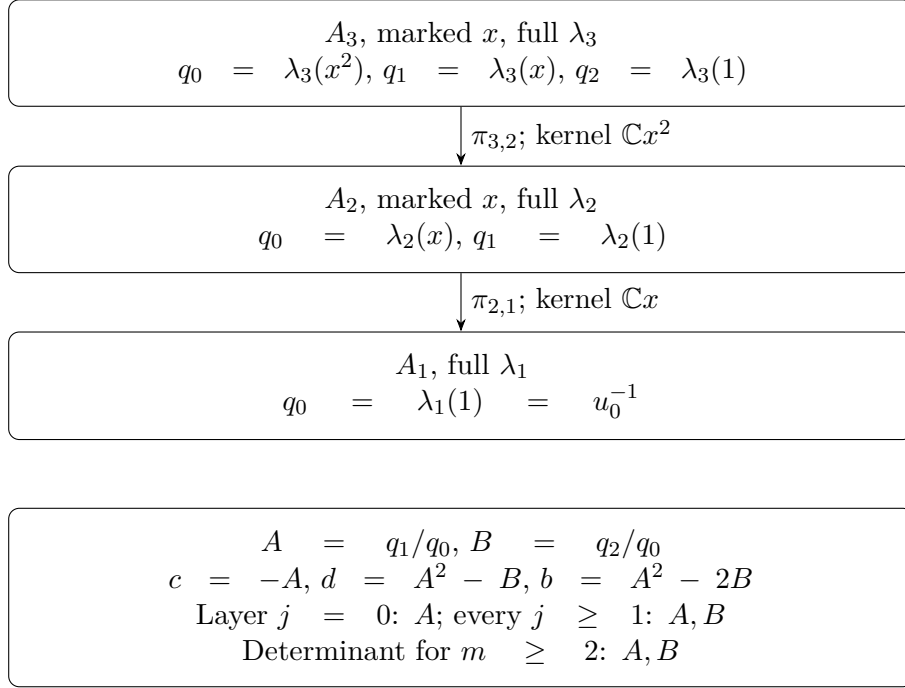


Figure 2: The coordinate-marked residue tower and the sharp fixed-test receivers. Each quotient's transpose under the perfect residue forms is multiplication by x , not an algebra inclusion. RJ3 proves these maps and their kernels; RJ6–9 prove all displayed receiver thresholds using the original t and $s = \rho + ix/2$. The energy receiver is credited to the supplied Theorem `thm:residuejet`; the fixed-test targets are HR4 and HR14. The diagram records analytic thickenings and does not assign a zero multiplicity.

3 The rational witness in the complete arithmetic trace

The supplied `rational_witness.json` is the exact vector defined in *A non-Eulerian RH counterexample attempt*, equations (4.2)–(7.2). We preserve that vector, its original coordinate Z , and the physical times $-2, 0$. This section proves its receiving formulas, reproduces its four reported signs independently, and calculates how the contact term BT3 would enter these same tests. The numerical calculation is a reproduction of the supplied result; it is not attributed as a new discovery. The contact receiver WR12–14 is an additional calculation. The two supplied source files are retained with this edition.

WR1. The original theta family and complete root trace. Use the original Rodgers–Tao family, with their equations `htdef`, `hoz`, `sas` in *The de Bruijn–Newman constant is non-negative*:

$$H_t(Z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, \quad \Phi(u) = \sum_{l \geq 1} (2\pi^2 l^4 e^{9u} - 3\pi l^2 e^{5u}) e^{-\pi l^2 e^{4u}}. \quad (\text{WR1})$$

Every summand is positive for $u \geq 0$, since $2\pi l^2 e^{4u} - 3 > 0$. On compact real time intervals, $\Phi(u) \leq C e^{9u} e^{-\pi e^{4u}/2}$ provides an integrable majorant after multiplication by any fixed power of u and any $e^{Tu^2 + Ru}$. Thus the integral is entire in Z , differentiable in t at every order, even, real on the real axis, and satisfies $\partial_t H = -H_{ZZ}$ and $H_t(0) > 0$. The same bound, maximizing $Ru - (\pi/8)e^{4u}$ after absorbing the fixed $Tu^2 + 9u$, gives $\log \max_{|Z| \leq R} |H_t(Z)| = O_T(R \log(R+2))$. Jensen’s formula then bounds its zero count by $O_T(R \log(R+2))$; in particular $\sum_\zeta m_\zeta |\zeta|^{-2} < \infty$. The order-one Hadamard factorization used in the supplied source (Jacques Hadamard, *Étude sur les propriétés des fonctions entières*, 1893, pp. 171–215), paired at $\zeta, -\zeta$, therefore gives near zero

$$\frac{H'_t(Z)}{H_t(Z)} = - \sum_{n \geq 1} S_n(t) Z^{2n-1}, \quad S_n(t) = \sum_{\zeta: H_t(\zeta)=0} m_\zeta \zeta^{-2n}. \quad (\text{WR2})$$

Both members of every pair are included. Indeed a paired factor is $1 - Z^2/\zeta^2$ and its logarithmic derivative is $-2 \sum_{n \geq 1} Z^{2n-1}/\zeta^{2n}$. The possible linear exponential in the factorization has zero slope because H_t is even. This verifies all constants in WR2.

Put $F^\dagger(Z) = \overline{F(\bar{Z})}$. For $f_c(Z) = \sum_{i=0}^{d-1} c_i Z^{-2i-1}$, define

$$Q_t(f_c, f_d) = \sum_\zeta m_\zeta \overline{f_c(\bar{\zeta})} f_d(\zeta), \quad K_d(t)_{ij} = S_{i+j+1}(t). \quad (\text{WR3})$$

No test has a pole at a root because $H_t(0) > 0$. The series is absolutely convergent by the preceding zero bound and the $O(|Z|^{-2})$ decay of the product. Expanding the finite test sums proves $Q_t(f_c, f_d) = c^* K_d(t) d$. Complex conjugation permutes the zeros, so K_d is real symmetric. This is the Jacobian-weighted trace of the complete local algebras: for $H_t = (Z - \zeta)^m u$, the local residue of $f_c^\dagger f_d H'_t / H_t$ is $\overline{m f_c(\bar{\zeta})} f_d(\zeta)$. The term u'/u is holomorphic, while every local algebra and its unit are retained before that specified contraction. Under $s = 1/2 + iZ/2$, this is exactly the original reflection $s \mapsto 1 - \bar{s}$; the residue Jacobian $i/2$ and logarithmic-derivative factor $-2i$ multiply to one. At a zero value of this form the supported scalar remains e , with the zero semilattice retained as in BT22.

WR2. Exact theta moment and heat production maps. Let $M_n(t) = \int_0^\infty u^{2n} e^{tu^2} \Phi(u) du$ and $a_n = (-1)^n M_n / (2n)!$. The preceding majorant permits termwise Taylor expansion, so $H_t = \sum a_n Z^{2n}$. Multiplying WR2 by this series and equating Z^{2n-1} proves

$$M_0 S_n = -2n a_n - \sum_{j=1}^{n-1} a_j S_{n-j}. \quad (\text{WR4})$$

The positive denominator $M_0 = H_t(0)$ is kept. Writing $R = H'/H$ and differentiating the PDE gives $\partial_t \log H = -(R' + R^2)$. Equating the coefficient of Z^{2n} in this identity proves the complete recurrence

$$S'_n = 2n \left[\sum_{a=1}^n S_a S_{n+1-a} - (2n+1)S_{n+1} \right]. \quad (\text{WR5})$$

All coefficients are analytic in real time, because $H_t(0) > 0$. These identities therefore hold through collisions elsewhere in the zero set without division by a zero gap. For a fixed vector c , its derivative is $c^* K'_d(t)c$, with every matrix entry given by WR5; there is no additional coefficient-motion term.

WR3. The exact supplied vector. For the nineteen integers n_i in the retained JSON, the original coefficients are

$$c_i = \frac{1000^i n_i}{10^{120}}, \quad 0 \leq i \leq 18, \quad n_{18} = 10^{120}, \quad c_{18} = 10^{54}. \quad (\text{WR6})$$

Thus the conditioning diagonal has an explicit invertible map $c = Dv$, $D_{ii} = 1000^i$; the actual function has not had its coordinate or amplitude changed. The entire vector is provided as exact integers rather than rounded coefficients. WR3–5 show that moments through 38 suffice for both $Q_*(t) = c^* K_{19}(t)c$ and $Q'_*(t)$.

The independent interval calculation in `reproduce_rational_witness.py` gives the following outward rational bounds, with all root contributions included:

$$\begin{aligned} Q_*(-2) &\in [-2.385, -2.383]10^{-46}, \\ Q'_*(-2) &\in [1.0766, 1.0767]10^{-44}, \\ Q_*(0) &\in [0.00081442298606666852, 0.00081442298606666854], \\ Q'_*(0) &\in [0.00218680919457417358, 0.00218680919457417359]. \end{aligned} \quad (\text{WR7})$$

The full binary endpoints are in `REPRODUCED_RATIONAL_WITNESS.json`. They overlap the supplied endpoints, and each independent enclosure has the asserted strict sign. The following calculation supplies the error bounds; reading signs from the supplied JSON alone was not used as a proof. No 32-dimensional inertia claim from the companion manuscript is certified by this nineteen-dimensional reproduction.

WR4. Bracketed quadrature and analytic error. Use 160-point Gauss–Legendre quadrature on each of the 24 panels of $[0, 3]$, with the first 16 summands in WR1, at 360 decimal digits in directed interval arithmetic. Every Legendre root is bracketed by rational decimal endpoints: the recurrence evaluates opposite strict signs at the endpoints; all 160 brackets are disjoint in $(-1, 1)$. Since the degree is 160, there is exactly one root in each bracket. The recurrence also encloses $P'_{160}(x) = 160(xP_{160}(x) - P_{159}(x))/(x^2 - 1)$ and the positive weight $2/((1 - x^2)P'_{160}(x)^2)$. Approximate Newton iterates are used only to propose brackets; the signs, disjointness and weight denominators are checked by the interval calculation.

For panel centre v , length $\ell = 1/8$, take the Bernstein ellipse of parameter 4, with semiaxes $A = 17/128$, $B = 15/128$. Let $x_- = v - A$, $x_+ = v + A$, $R^2 = x_+^2 + B^2$ and $\kappa = 1 - (4B)^2/2 > 0$. Then $|z|^2 \leq R^2$ and $\cos(4\Im z) \geq \kappa$ on the ellipse. Since $3 < \pi < 4$, the first sixteen theta terms have absolute sum at most

$$P_v = \left(32e^{9x_+} \sum_{l=1}^{16} l^4 + 12e^{5x_+} \sum_{l=1}^{16} l^2 \right) e^{-3e^{4x_+} \kappa}. \quad (\text{WR8})$$

At the two times used here, $|e^{tz^2}| \leq e^{|t|B^2}$. The moment- k integrand is therefore bounded by $P_v e^{|t|B^2} R^{2k}$. Pullback to the annulus by $(w + w^{-1})/2$ and the Cauchy coefficient estimate bound

the Chebyshev tail after degree 319 by $2M4^{-319}/3$. Gauss quadrature is exact on that polynomial, and its positive weights sum to ℓ . Bounding both the integral and quadrature of the tail proves the panel error

$$\frac{4\ell}{3}P_v e^{|t|B^2} R^{2k} 4^{-319}. \quad (\text{WR9})$$

WR5. The two omitted tails. On $[0, 3]$, the omitted positive summands satisfy $\Phi_{>16}(u) \leq 32e^{27} \sum_{l \geq 17} l^4 e^{-3l^2}$. The successive ratio is at most $16e^{-105} < 1/2$. At $t = -2, 0$ the complete omitted integral is consequently at most

$$192 \cdot 17^4 \cdot 3^{2k} e^{-840}. \quad (\text{WR10})$$

For $u \geq 3$, the same ratio bound starting at $l = 1$ gives $\Phi(u) \leq 64e^{9u} e^{-3e^{4u}}$. For $0 \leq k \leq 38$ and $t \leq 0$, $2k \log u + 9u \leq 85u \leq 29u^2 \leq e^{4u}$. The last inequality holds at 3 and persists because e^{4u}/u^2 is increasing. Also $e^{4u} \geq 1000(1 + 4(u - 3))$. Thus the spatial tail is less than

$$64 \int_3^\infty e^{-2e^{4u}} du \leq (64/8000)e^{-2000} < e^{-2000}. \quad (\text{WR11})$$

Adding WR9 over all panels and WR10–11 encloses the full moment. The code then applies WR4 and WR5 in outward intervals with exact factorials and exact rational coefficients. Every division by M_0 is preceded by a positive-interval check. Finite interval arithmetic and induction over these recurrences give enclosures of the actual S_n, S'_n and hence WR7. This is computer-assisted interval verification using `mpmath.iv`; it does not claim formal verification of that library. Binary endpoints, precision and every analytic remainder are retained in the reproducible calculation.

WR6. The additional contact receiver for the same Laurent tests. At any nonzero repeated centre c in BT1–6, with multiplicity m , put $M = m(m-1)$. Taking $\phi(Z) = Z^{-2i-1} Z^{-2j-1}$ in BT8–9 gives the exact matrix

$$(\delta_{\text{contact}} \dot{K})_{ij} = -M 2(i+j+1)(2i+2j+3)c^{-2i-2j-4}. \quad (\text{WR12})$$

Sum this expression over all retained contact centres. It uses the actual coefficient map WR6; in particular the contribution to this very witness is

$$\delta_{\text{contact}} \dot{Q}_* = - \sum_l m_l(m_l-1)(f_*^\dagger f_*)''(c_l). \quad (\text{WR13})$$

Twice differentiating the monomials proves WR12, and bilinearity proves WR13. At a real centre where $f_*(c) = 0$, its single-centre value is $-2m(m-1)|f'_*(c)|^2$. The term includes the second test derivatives as well when the value does not vanish.

For an even finite polynomial P evolving under BT2 with $U = 1$ and $P(t, 0) \neq 0$, its inverse-root moments obey the entire forced analogue of WR5 almost everywhere, and distributionally on compact time intervals:

$$S'_n = 2n \left[\sum_{a=1}^n S_a S_{n+1-a} - (2n+1)S_{n+1} \right] - 2n(2n+1) \sum_l \frac{m_l(m_l-1)}{c_l^{2n+2}}. \quad (\text{WR14})$$

Indeed $\partial_t \log P = -(R' + R^2) + D/P$ and BT5 gives $D/P = \sum_l M_l/(Z - c_l)^2$. Its coefficient at Z^{2n} is $(2n+1) \sum_l M_l c_l^{-2n-2}$. Comparing that coefficient with $\partial_t \log P$ proves WR14 with its minus sign. For a nonconstant unit BT3 retains the additional holomorphic term B ; it may not be dropped in this coefficient comparison. These formulas identify exactly which arithmetic observations would detect residence at a collision. They do not assert such residence in the original analytic theta flow or replace its genuine recurrence WR5. WR7 is evaluated using WR5 throughout.